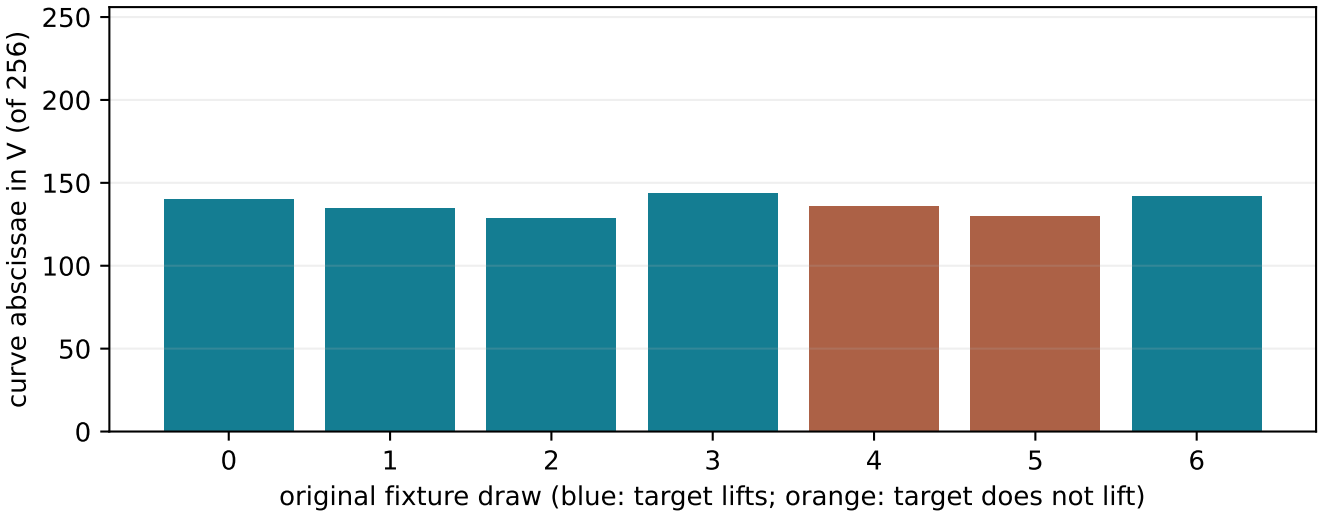
Original fixtures: finite curve-membership audit

Diagnostic date 2026-10-25. Source frozen at c470f5d51 before output. The byte-identical original seven Boolean fixtures use GF(2^{22}), modulus $X^{22} + X + 1$, and $E_0 : y^2 + xy = x^3 + 1$. Exact input SHA-256 (two joined lines):
`95ee147e1221ce4ef6c98bdd0b808398`
`a01362343ffaa66debc217d6eeff5a9b`.



Draw	Boolean solutions	target on curve	curve x in V	curve points in V
0	0	yes	140	279
1	6	yes	135	269
2	6	yes	129	257
3	0	yes	144	287
4	18	no	136	271
5	0	no	130	259
6	0	yes	142	283

The native audit finds 3/4 UNSAT targets and 2/3 SAT targets liftable. Draws 4 and 5 are not ordinary curve-query abscissae for this model. Draw 0, used by the all-eight balanced k=3 replay, is liftable. This qualifies the application of these fixtures; it does not invalidate their finite Boolean-system solution counts or refutation observations.

Every returned y root is checked by direct curve-equation substitution. An independently evaluated Artin–Schreier trace criterion agrees for every target and every element of each complete 256-element V. Counts include x=0 and its two-torsion point. Neither target nor factor-base subgroup membership is established. Curve membership alone does not prove a signed point decomposition, a degree law, a complete DLP solve or a speed result.

Execution: native Rust, local shared x86-64 host; 12 GiB address-space cap, 900-second limit, exit 0, no timing comparison. A separate exhaustive small-field test compares every y value at n=5,6,7; all three example tests pass. Protocol: `../../FIXTURE_GROUP_PROTOCOL.md`. Source: `examples/ic_fixture_group_audit.rs`. Raw receipt and provenance: `../../pipeline-receipts/fixture-group-audit-20261025/`. The old branch reports and source-pinned inputs are preserved unchanged.